
ME 526 : Programming Paradigm for Open-Source Software

Course Code : ME 526

Course Name : Programming Paradigm for Open-source Software

L-T-P-C : 2-0-2-3

Intended for : UG

Prerequisite : ME 303 - Heat Transfer & ME - 213 Engineering Thermodynamics, ME - 210 Fluid Mechanics

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Language**

- Module 1: Object oriented fundamentals, C++ Basics, C++ Functions (6 Hours)
- Module 2: Encapsulation, Constructors and Destructors, Function and Programming Overloading. (7 Hours)
- Module 3: Inheritance, Virtual functions, Dynamic binding (5 Hours)
- Module 4: Containment and Private Inheritance, Designing with C++: Template and exceptions. (5 Hours)

- **Debugger**

- Module 5: Navigating Through Codes, Stopping and Running the Program, Examining data (3 Hours)

- **Introduction to Open-Source Software**

- Module 6: File and folder systems, make and makefile, including a file to open-source software, correction of a bug (2 Hours)

Textbooks:

1. Bjarne Stroustrup, **The C++ Programming Language**, 4th Edition, Addison Wesley Professional, 2013.
2. Andreas Zeller et al., **Debugging with DOD**, User's Guide and Reference Manual 2000.

References:

1. Stanley B. Lippman, Josee Lajoie, and Barbara E Moo, **C++ Primer**, 5th Edition, Addison-Wisley Professional, 2012.
2. Steve Oualline, **Practical C++ Programming**, 2nd Edition, O'Reilly Media, 2002
3. Bjarne Stroustrup, **Programming: Principles and Practice Using C++**, Addison-Wisley Professional 2008.
4. OpenFOAM: User Guide V21 12